

Product Requirements Document (PRD): Defensible Compliance Shield

Status: DRAFT (Sections 1-3) **Date:** 2026-01-28 **Source Material:** Procurement Officer JTBD Interview

1. Job To Be Done (JTBD)

When I am approving a procurement contract in a highly regulated aviation environment, I want to be confident that no critical compliance, contractual, or sanctions-related detail has been missed, so that I can sign the contract with certainty, protect my professional reputation, and avoid irreversible audit or regulatory consequences — even if the approval process takes longer.

2. User Persona

Role: Senior Procurement Officer. **Context:** Works in a high-stakes, regulated environment (aviation) where personal accountability for compliance is high. **Mindset:** Driven by risk aversion and self-preservation ("sleeping at night") rather than operational speed. **Environment:** Overwhelmed by volume, managing disjointed systems (Word, PDF, ERP, Sanctions Lists), and subject to strict internal and external audits.

3. Current Pain Points

- Manual Reconciliation Fatigue:** Users must manually cross-reference data across multiple disconnected sources (vendor PDFs vs. internal templates, pricing vs. budget, sanctions lists vs. vendor names), increasing cognitive load and the likelihood of human error.
- High Personal Liability:** The "audit trail ends at the signature." Officers feel personally exposed to career-ending risks (sanctions violations, liability gaps) if they miss a detail while rushing, creating a "gambling" dynamic with every approval.
- Lack of Trusted "Source of Truth":** Existing systems are rigid and often contain stale data. Users are forced to maintain personal "shadow ledgers" (spreadsheets) to track the actual status of contracts, as the official system is viewed merely as a compliance hurdle rather than a helpful tool.
- Assymmetric Incentives:** The upside of speed (happy operations) is minimal compared to the catastrophic downside of a compliance error (audit findings, termination), forcing users to deprioritize efficiency in favor of manual, defensive double-checking.

4. Functional Requirements

The system must provide a "Defensibility Shield" that augments the officer's judgment with verifiable evidence, rather than replacing it with automation.

4.1. Intelligent Discrepancy Highlighting

- Requirement:** The system must visually compare incoming supplier documents (PDFs, Word) against the approved internal template or policy versions.
- Feature:** "Redline Warning Overlay" - highlights deviations in key clauses (e.g., Indemnity Caps, Payment Terms, Anti-Corruption) directly on the document viewer.
- Goal:** Direct the user's attention to high-risk changes without deciding if they are acceptable.

4.2. Point-in-Time Compliance Verification

- **Requirement:** The system must perform real-time checks against external and internal databases (Sanctions lists, Tax Clearance status, Internal Vendor Master) *at the moment of review*.
- **Feature:** "Compliance Snapshot" - a downloadable, timestamped report generated with a single click that proves the vendor was compliant at that specific date and time.
- **Goal:** Provide an irreversible audit artifact ("The Shield") that the officer can attach to the approval.

4.3. Interactive Compliance Checklist

- **Requirement:** The system must present a dynamic checklist based on the contract value and category (e.g., "Software" vs "Aircraft Spare Parts").
- **Feature:** "Guided Verification Flow" - requires the user to explicitly acknowledge critical checks (e.g., "I have verified the ISO certificate expiry date") rather than just clicking "Approve."
- **Goal:** Force cognitive engagement with critical risks and create a record of specific user affirmations.

4.4. Integrated "Source of Truth" Dashboard

- **Requirement:** The system must aggregate relevant data points (Budget status, Vendor Risk Rating, Contract Expiry) from disparate backend systems into a single view alongside the contract.
- **Feature:** "Decision Context Panel" - displays key metadata next to the document viewer, eliminating the need to alt-tab between ERP and other portals.
- **Goal:** Reduce manual reconciliation fatigue while ensuring the user sees the complete picture before signing.

4.5. Audit Trail of Review Actions

- **Requirement:** The system must log not just the final approval, but the *process* of review.
- **Feature:** "Review Activity Log" - records that the user viewed the "Compliance Snapshot," opened the "Redline Overlay," and completed the checklist.
- **Goal:** Prove due diligence was performed, protecting the officer from accusations of negligence in future audits.

5. Non-Functional Requirements

The system must operate within the strict constraints of a national carrier under regulatory scrutiny.

5.1. Data Residency & Sovereignty

- **Requirement:** All transactional data and audit logs must be physically stored within the Republic of Kazakhstan in compliance with local data protection laws (Law No. 94-V on Personal Data and Protection).

- **Constraint:** No sensitive contractual data may be processed by public cloud LLMs hosted outside the agreed jurisdiction without explicit legal clearance.

5.2. Auditability & Immutable Logging

- **Requirement:** All system actions (approvals, view events, document uploads) must be recorded in WORM (Write Once, Read Many) storage.
- **Constraint:** Audit logs must be retained for a minimum of 5 years to cover the standard tax audit window.
- **Access:** External auditors must be able to view a "ReadOnly" version of the decision trail without operational access.

5.3. Explainability of automated Flags

- **Requirement:** Any "Redline Warning" or "Compliance Flag" generated by the system must be traceable to a specific source document or policy rule.
- **Constraint:** "Black Box" AI recommendations are strictly forbidden. The system must cite the specific clause page number, sanctions list entry, or internal policy paragraph that triggered the alert.

5.4. Security & Access Control

- **Requirement:** Role-Based Access Control (RBAC) must mirror the internal Delegation of Authority (DoA) matrix exactly.
- **Constraint:** Approval limits must be hard-coded based on the officer's grade; the system must not allow a "Senior Officer" to approve a contract above their financial threshold.

5.5. Integration Latency

- **Requirement:** Compliance checks against external databases (Sanctions, Tax) must complete in under 30 seconds.
- **Constraint:** The system must degrade gracefully; if an external registry is down, the system must block approval until the check can be performed, rather than allowing a "bypass."

6. Business Case & Decision

6.1. Financial Assessment (ROI)

The proposed system is expected to save approximately **6 hours per week per Senior Procurement Officer** by reducing manual reconciliation and rework, while preserving human judgment.

- **Annual Productivity Gain:** ~1.7 million KZT per employee.
- **Total Annual Benefit (Team of 10):** >17 million KZT.
- *Note:* This excludes value from avoided compliance penalties and reduced audit escalations.

6.2. Governance & Ethical Assessment

The solution is designed to operate fully within the regulatory constraints of the Republic of Kazakhstan:

- **Data Sovereignty:** All sensitive data and audit logs are stored domestically.
- **Security:** Strict role-based access controls and immutable audit trails.
- **Reliability:** The system does NOT make autonomous decisions. All flags are fully explainable and interpretable, linked to specific source documents and policies, mitigating hallucination and compliance risks.

6.3. Executive Decision

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Rationale: This initiative addresses a clearly validated internal pain point rooted in personal accountability and compliance risk, not superficial efficiency gains. The solution delivers strong defensive value with manageable governance risks and a positive financial impact, while aligning with Kazakhstan's AI ethics principles and regulatory requirements.